1 sets

a **set** is an unordered collection of objects. each set can contain items of mixed types, and it can also contain other sets. the objects that make up the set are called **elements** of the set.

1.1 describing a set

there are three common ways to describe a set:

- explicit enumeration
  - $A = \{1, 2, 3, 4\}$
- ellipses if the pattern is trivial
  - $B = \{2, 4, 6, ..., 12\}$
- set builder notation
  - $C = \{y \mid y = 3k \text{ for some integer } k\}$
  - plain english: "this set C contains all elements y such that $y = 3k$ for some integer k."

1.2 notable sets

these are some common sets that are predefined and commonly used in mathematics:

$$\begin{array}{ll}
\mathbb{N} = \{0, 1, 2, 3, ...\} & \text{natural numbers} \\
\mathbb{Z} = \{..., -2, -1, 0, 1, 2, ...\} & \text{integers} \\
\mathbb{Z}^+ = \{1, 2, 3, ...\} & \text{positive integers} \\
\mathbb{Q} = \left\{\frac{p}{q} \mid p, q \in \mathbb{Z}, q \neq 0\right\} & \text{rational numbers} \\
\mathbb{R} & \text{real numbers} \\
\emptyset = \{\} & \text{empty set}
\end{array}$$

1.3 set equality

two sets are *equal* if and only if they contain exactly the same elements (excluding duplicated elements).

$$A = B \text{ iff } \forall x \, (x \in A \leftrightarrow x \in B)$$

1.4 subsets

some set A is a **subset** of another set **B** if and only if every element of A is an element in the set B. we write this as $A \subseteq B$, and call B a **superset** of A.

$$A \subseteq B \text{ iff } \forall x \, (x \in A \to x \in B)$$

1.4.1 proper subset

some A is a **proper subset** of B if and only if $A \subseteq B$ but $A \neq B$. we write this as $A \subset B$.

$$A \subset B \text{ iff } \forall x\,(x \in A \to x \in B) \land \exists y\,(y \in B \land y \notin A)$$

1.4.2 subset properties

- for all sets S, $\emptyset \subseteq S$.
- for any set S, $S \subseteq S$.
- if $S_1 = S_2$, then $S_1 \subseteq S_2$ and $S_2 \subseteq S_1$.

1.5 set operations

we can perform operations between sets as well.

1.5.1 $\cup$ union

the **union** of two sets A and B contains every element that is either in A or in B. we write this as $A \cup B$.

$$A \cup B = \{x \mid x \in A \lor x \in B\}$$

1.5.2 $\cap$ intersection

the **intersection** of two sets A and B contains every element that is in A and also in B. we write this as $A \cap B$.

$$A \cap B = \{x \mid x \in A \land x \in B\}$$

we say that two sets are **disjoint** if $A \cap B = \emptyset$.

1.5.3 $-$ difference

the **difference** of two sets A and B contains every element that is in A but not in B. we write this as $A - B$.

$$A - B = \{x \mid x \in A \land x \notin B\}$$

1.5.4 $\bar{A}$ complement

the complement of a set A, denoted by $\bar{A}$, contains every element that is in the universal set U, but not in A.

$$\bar{A} = \{x \mid x \in U \land x \notin A\}$$

1.5.5 $|S|$ cardinality

let S be a set. if there are exactly n elements in S, where n is a non-negative integer, then S is a finite set whose cardinality is n. the cardinality of S is denoted by $|S|$.

$$S = \{1, 2, 3, 4, 5\}$$
$$|S| = 5$$

1.6 power set

given a set S, its power set is the set containing all subsets of S. we denote the power set of S as $P(S)$.

example:

$$P(\{1\}) = \{\emptyset, \{1\}\}$$
$$P(\{1, 2, 3\}) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$$

note:

- the set $\emptyset$ is in the power set of any set.
  - $\forall S\,(\emptyset \in P(S))$
- the set S is in its own power set.
  - $\forall S(S \in P(S))$
- the cardinality of the power set of any set S is 2 to the power of the cardinality of set S.
  - $|P(S)| = 2^{|S|}$